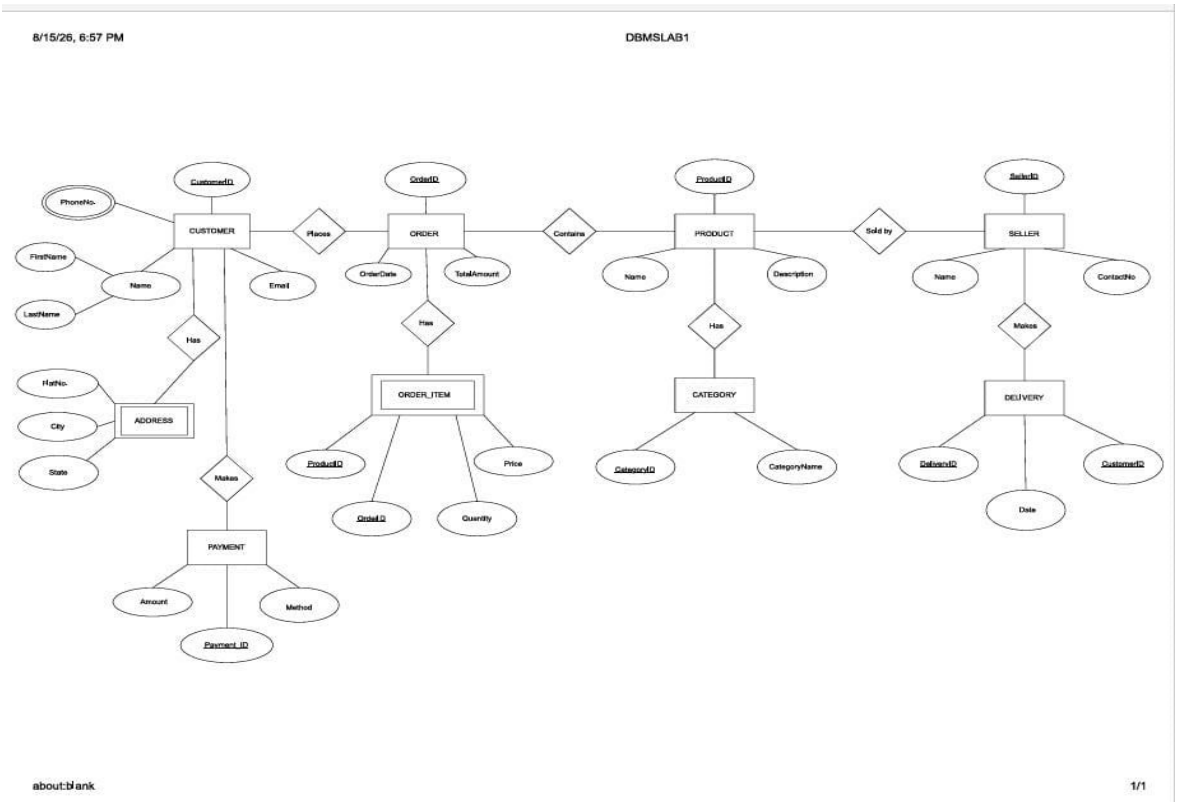
Experiment: 1

Aim:
To design an ER diagram for an Indian e-commerce platform by identifying entities, attributes, relationships, keys, and participation constraints.

Theory:
An Entity-Relationship (ER) diagram is a graphical representation of a database. It shows entities, their attributes, and relationships between them.
In an e-commerce system, entities such as Customer, Product, Order, Seller, Category, Payment, Delivery, Address, and OrderItem are connected through different relationships.

Key Points
- Primary Key: Uniquely identifies an entity.
- Composite Attribute: Attribute divided into smaller parts, e.g., Address → City, State, PIN.
- Multi-valued Attribute: Can have multiple values, e.g., Phone Numbers.
- Weak Entity: Depends on another entity for identification, e.g., OrderItem.
- Specialization: Divides an entity into sub-entities, e.g., Product → Physical Product, Digital Product.
- Cardinality: Shows relationship as 1:1, 1:N, or M:N.
- Participation: Can be total or partial.

ER Diagram:



Result:

The ER diagram for the e-commerce platform was successfully designed with the required entities, attributes, relationships, keys, and constraints.